

BANK OF COBOL

Core Banking System — Business Rules

Migration Reference Document

Source Program:	BANKING-SYSTEM (BANKING.cbl)
Platform:	IBM Mainframe (COBOL)
Prepared for:	System Migration — Business Analyst Review
Operations:	Deposit and Withdrawal only

Table of Contents

- 1. Account Data Model
- 2. Account Opening & Initialization
- 3. Account Status Management
- 4. Account Tier Classification (Regular vs VIP)
- 5. Deposit Processing
- 6. Withdrawal Processing
- 7. Transaction Limits Reference Table
- 8. Account Tier Promotion & Demotion
- 9. Transaction Logging & Audit Trail
- 10. End-of-Day Processing
- 11. Inquiries & Reporting
- 12. System Capacity & Constraints
- 13. Migration Notes for Business Analysts

1. Account Data Model

Each client account holds the following information:

Field	Description
Client ID	Unique 6-digit number assigned at account opening (starts at 100001)
Last Name	Up to 15 characters
First Name	Up to 10 characters
Balance	Current account balance
Status	ACTIVE, FROZEN, or CLOSED

Field	Description
Account Type	REGULAR or VIP
Daily Withdrawal Total	Cumulative withdrawals processed today (resets at end of day)
Daily Deposit Total	Cumulative deposits processed today (resets at end of day)
Date Opened	The date the account was opened
Total Transaction Count	Lifetime count of all transactions (approved and rejected)
Last Transaction Date	Date of the most recent transaction attempt
Last Transaction Type	Whether the last transaction was a Deposit or Withdrawal
Last Transaction Amount	Dollar amount of the most recent transaction

2. Account Opening & Initialization

BR-001 — Client ID Assignment
Every new account is assigned a unique 6-digit Client ID. IDs are assigned sequentially, beginning at 100001.

BR-002 — Opening Balance
All new accounts are funded with an initial balance of \$500.00 at the time of opening.

BR-003 — Account Type at Opening
At the time of opening, the system evaluates the account balance to determine account type: if the opening balance exceeds \$50,000.00, the account is classified as VIP; otherwise Regular.
Note: With a standard opening balance of \$500.00, all new accounts open as Regular.

BR-004 — Daily Accumulators at Opening
Daily withdrawal and deposit totals are initialized to \$0.00 at account opening.

3. Account Status Management

An account can be in one of three statuses. Status governs whether transactions are permitted.

Status	Code	Description
Active	A	Account is in good standing. All permitted transactions can be processed.
Frozen	F	Account has been suspended. No deposits or withdrawals are allowed. Customer must visit a branch to unfreeze.
Closed	C	Account is permanently closed. No transactions of any kind are permitted.

BR-005 — Frozen Account Restriction
No transaction (deposit or withdrawal) may be processed on a Frozen account.
"ACCOUNT FROZEN - VISIT BRANCH TO UNFREEZE"

BR-006 — Closed Account Restriction

No transaction may be processed on a Closed account.

"ACCOUNT IS PERMANENTLY CLOSED"

BR-007 — Status Check is First Priority

Account status is always validated before any amount or limit checks. If an account is Frozen or Closed, the transaction is immediately rejected with no further validation.

4. Account Tier Classification (Regular vs VIP)

BR-008 — VIP Qualification Threshold

An account qualifies for VIP status when the account balance exceeds \$50,000.00.

BR-009 — Regular Account

Any account with a balance of \$50,000.00 or less is classified as Regular and is subject to Regular transaction limits.

BR-010 — VIP Account

Any account with a balance above \$50,000.00 is classified as VIP and is subject to VIP transaction limits (significantly higher).

5. Deposit Processing

Deposits are validated in the following sequence. The first failure immediately rejects the transaction.

5.1 Deposit Validation Sequence

Step 1 — Account Status Check

- If account is FROZEN → Reject (see BR-005)
- If account is CLOSED → Reject (see BR-006)

Step 2 — Minimum Amount Check**BR-011 — Minimum Deposit Amount**

The minimum deposit amount is \$1.00.

"AMOUNT BELOW MINIMUM DEPOSIT OF \$1.00"

Step 3 — Single Transaction Cap**BR-012 — Regular Account: Single Deposit Limit**

A Regular account may not receive more than \$50,000.00 in a single deposit.

"EXCEEDS REGULAR SINGLE DEPOSIT CAP \$50,000"

BR-013 — VIP Account: Single Deposit Limit

A VIP account may not receive more than \$250,000.00 in a single deposit.

"EXCEEDS VIP SINGLE DEPOSIT CAP \$250,000.00"

Step 4 — Daily Cumulative Cap

BR-014 — Regular Account: Daily Deposit Limit

The total of all deposits to a Regular account in a single business day may not exceed \$100,000.00.

"DAILY DEPOSIT LIMIT REACHED \$100,000.00"

BR-015 — VIP Account: Daily Deposit Limit

The total of all deposits to a VIP account in a single business day may not exceed \$500,000.00.

"VIP DAILY DEPOSIT LIMIT REACHED \$500,000"

5.2 Approved Deposit Processing

When a deposit passes all validations, the system:

- Adds the deposit amount to the account balance.
- Adds the deposit amount to the account's daily deposit accumulator.
- Increments the transaction counter for the account by 1.
- Records the last transaction date, type (Deposit), and amount on the account.
- Evaluates whether the new balance qualifies the account for VIP upgrade (see BR-016).

6. Withdrawal Processing

Withdrawals are validated in the following sequence. The first failure immediately rejects the transaction.

6.1 Withdrawal Validation Sequence

Step 1 — Account Status Check

- If account is FROZEN → Reject (see BR-005)
- If account is CLOSED → Reject (see BR-006)

Step 2 — Minimum Amount Check

BR-020 — Minimum Withdrawal Amount

The minimum withdrawal amount is \$1.00.

"AMOUNT BELOW MINIMUM WITHDRAWAL OF \$1.00"

Step 3 — Single Transaction Cap

BR-021 — Regular Account: Single Withdrawal Limit

A Regular account may not withdraw more than \$10,000.00 in a single transaction.

"EXCEEDS REGULAR SINGLE WITHDRAWAL CAP \$10K"

BR-022 — VIP Account: Single Withdrawal Limit

A VIP account may not withdraw more than \$50,000.00 in a single transaction.

"EXCEEDS VIP SINGLE WITHDRAWAL CAP \$50,000"

Step 4 — Daily Cumulative Cap

BR-023 — Regular Account: Daily Withdrawal Limit

The total of all withdrawals from a Regular account in a single business day may not exceed \$20,000.00.
"DAILY WITHDRAWAL LIMIT REACHED \$20,000.00"

BR-024 — VIP Account: Daily Withdrawal Limit

The total of all withdrawals from a VIP account in a single business day may not exceed \$100,000.00.
"VIP DAILY WITHDRAWAL LIMIT REACHED \$100,000"

Step 5 — Balance Adequacy Checks

BR-025 — Minimum Required Balance

After a withdrawal, the account balance must not fall below \$100.00.
"WOULD BREACH MINIMUM REQUIRED BALANCE \$100.00"

BR-026 — No Overdraft

The system does not permit overdrafts. A negative post-withdrawal balance is not allowed.
"INSUFFICIENT FUNDS - OVERDRAFT NOT PERMITTED"
Note: BR-025 is checked first. BR-026 is only reached if the minimum balance were \$0.00.

6.2 Approved Withdrawal Processing

When a withdrawal passes all validations, the system:

- Deducts the withdrawal amount from the account balance.
- Adds the withdrawal amount to the account's daily withdrawal accumulator.
- Increments the transaction counter for the account by 1.
- Records the last transaction date, type (Withdrawal), and amount on the account.
- Evaluates whether the new balance requires VIP → Regular downgrade (see BR-017).

7. Transaction Limits Reference Table

Limit	Regular Account	VIP Account
Minimum transaction amount	\$1.00	\$1.00
Maximum single deposit	\$50,000.00	\$250,000.00
Maximum single withdrawal	\$10,000.00	\$50,000.00
Maximum daily deposits (total)	\$100,000.00	\$500,000.00
Maximum daily withdrawals (total)	\$20,000.00	\$100,000.00
Minimum required balance	\$100.00	\$100.00
Overdraft permitted	No	No

8. Account Tier Promotion & Demotion

BR-016 — Automatic VIP Upgrade After Deposit

After a deposit is applied, if the new balance exceeds \$50,000.00 and the account is currently Regular, the account is automatically upgraded to VIP. New VIP limits apply immediately.

BR-017 — Automatic VIP Downgrade After Withdrawal

After a withdrawal is applied, if the new balance is \$50,000.00 or less and the account is currently VIP, the account is automatically downgraded to Regular. The operator is notified: "NOTE: ACCOUNT DOWNGRADED VIP -> REGULAR".

BR-018 — Tier Applies at Time of Transaction

The account type at the moment of transaction submission determines which limits apply. A mid-day upgrade or downgrade takes effect immediately for all subsequent transactions.

9. Transaction Logging & Audit Trail

BR-030 — All Transactions Are Logged

Every transaction attempt — whether approved or rejected — is recorded in the transaction log. A rejected transaction is not applied to the balance but is still written to the log.

BR-031 — Transaction Log Contents. Each log entry captures:

- Transaction ID (sequential, system-generated)
- Transaction Date and Time
- Client ID
- Transaction Type (Deposit or Withdrawal)
- Amount
- Account Balance after the transaction (approved transactions)
- Result: Approved (OK) or Rejected (RJ)
- Rejection Reason (if rejected)

BR-032 — Circular Log Buffer

The transaction log holds a maximum of 5,000 entries. When full, new entries overwrite the oldest entries (circular buffer). The total count is retained across the session.

BR-033 — Transaction IDs are Sequential

Each log entry is assigned a unique, sequentially incrementing Transaction ID starting at 1. IDs are never reused within a session.

10. End-of-Day Processing

BR-040 — Daily Limit Reset

At the end of each business day, an authorized operator must manually trigger the end-of-day reset. This clears daily withdrawal and deposit accumulators to \$0.00 for every account and marks the beginning of a new business day.

BR-041 — Confirmation Required

The end-of-day reset requires explicit operator confirmation (Y/N prompt) before executing. If the operator does not confirm, the reset is cancelled and no data is changed.

BR-042 — Reset Applies to All Accounts

The daily reset applies simultaneously to all accounts regardless of status or type. Frozen and Closed accounts also have their daily accumulators cleared.

BR-043 — Daily Limits are Per Business Day

Transaction limits are calculated from the start of the current business day. After an end-of-day reset, every account starts the new day with \$0.00 accumulated toward daily limits.

11. Inquiries & Reporting

11.1 Balance Inquiry (BR-050)

Any account (including Frozen and Closed) may be queried for its current balance. Displays:

- Current balance
- Total amount withdrawn today
- Total amount deposited today
- Lifetime transaction count

11.2 Account Detail (BR-051)

A full account record view displays all account fields, plus the applicable transaction limits based on current account type (Regular or VIP), and the minimum required balance.

11.3 Client List (BR-052)

The client list is displayed 20 records at a time. The operator selects a starting record number (1 to 1000). Shows Client ID, name, balance, status, and account type for each account.

11.4 Daily Summary Report (BR-053)

The daily summary provides an aggregate view for the current business day:

- Total deposits across all accounts today
- Total withdrawals across all accounts today
- Total number of transactions logged in the session
- Count of accounts by status: Active, Frozen, Closed

11.5 Transaction Log View (BR-054)

The operator can view the most recent 20 transactions. Entries show: Transaction ID, date, Client ID, transaction type, amount, and result (OK/RJ).

12. System Capacity & Constraints

Parameter	Value
Maximum accounts in database	1,000
Maximum transaction log entries	5,000 (circular — oldest overwritten)
First assigned Client ID	100001
Supported transaction types	Deposit and Withdrawal only
Overdraft facility	Not supported
Inter-account transfers	Not supported
Interest calculation	Not supported
Account creation via system	Not supported (pre-loaded database)
Account closure via system	Not supported (status is pre-set)

13. Migration Notes for Business Analysts

The following values are hardcoded in the COBOL source and must be externalized as configurable parameters in the new system:

Rule Ref	Parameter	Current Value	COBOL Variable
BR-025	Minimum required balance	\$100.00	BR-MIN-BALANCE
BR-021	Regular max single withdrawal	\$10,000.00	BR-MAX-WD-SINGLE
BR-012	Regular max single deposit	\$50,000.00	BR-MAX-DEP-SINGLE
BR-023	Regular max daily withdrawal	\$20,000.00	BR-MAX-WD-DAILY
BR-014	Regular max daily deposit	\$100,000.00	BR-MAX-DEP-DAILY
BR-011/020	Minimum transaction amount	\$1.00	BR-MIN-TX-AMOUNT
BR-002	Initial account balance	\$500.00	BR-INITIAL-BALANCE
BR-008	VIP qualification threshold	\$50,000.00	BR-VIP-THRESHOLD
BR-022	VIP max single withdrawal	\$50,000.00	BR-VIP-MAX-WD-SINGLE
BR-013	VIP max single deposit	\$250,000.00	BR-VIP-MAX-DEP-SINGLE
BR-024	VIP max daily withdrawal	\$100,000.00	BR-VIP-MAX-WD-DAILY
BR-015	VIP max daily deposit	\$500,000.00	BR-VIP-MAX-DEP-DAILY

Key Design Decisions to Resolve During Migration

1. **VIP threshold logic is balance-based only.** There is no minimum tenure, credit score, or manual approval pathway — VIP status is granted and revoked automatically based solely on the balance crossing \$50,000.

2. **End-of-day reset is manual.** The current system requires a human operator to trigger the daily limit reset. The new system must decide whether this is automated (scheduled batch job) or remains operator-initiated.
3. **Rejected transactions are fully logged.** The audit trail includes both approved and rejected attempts. This behaviour must be preserved in the new system.
4. **No overdraft product exists.** The current system has no concept of an authorized overdraft. If the new system introduces overdraft, new business rules will be required.
5. **Frozen vs. Closed distinction.** Frozen is a temporary, reversible state that requires a branch visit to lift. Closed is permanent. The new system must preserve this distinction and implement the branch-visit workflow for unfreezing accounts.